Microalgal RNA-seq analyses to identify enzymes involved in the synthesis of bioactive compounds

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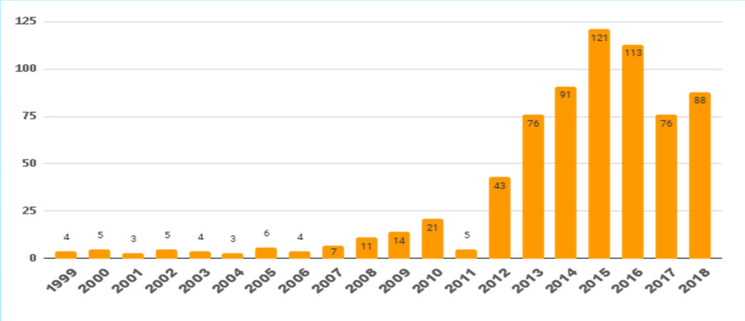
1. Introduction

Enzymes play vital roles in the performance of many biotechnological and industrial applications. **Microalgae** are very promising source of novel enzymes since they can be cultivated in photobioreactors, allowing the production of high biomass levels with a relatively cost-efficient and environment-friendly outcome. The growing interest in this field is well proved by the increasing number of related papers *[Vingiani et al. Marine Drugs, 2019]*.

With recent advances in microalgal **genomics and transcriptomic approaches**, it is becoming easier to identify sequences encoding targeted enzymes, increasing the likelihood of the identification, heterologous expression, and characterization of these enzymes of interest.



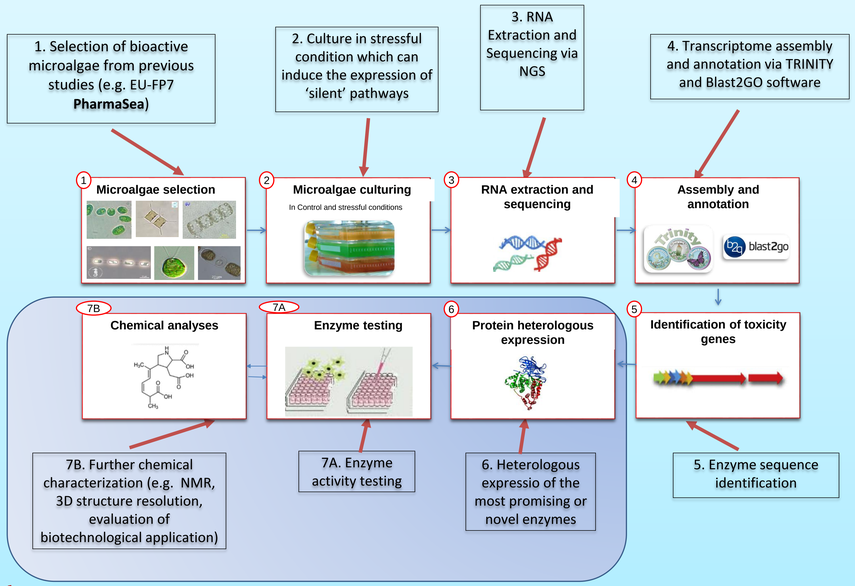
N. of Papers Related to Microalgae Biotechnology in the 20-years interval 1999–2018 (PubMed)

2. Pipeline for Enzyme Discovery in *Alexandrium tamutum*

Global Warming is causing the increase of occurrence of different **toxic algal blooms**, that cause severe damages to human health and economics. Collecting more data on the identity and synthesis of microalgal toxins is a key asset to predict these phenomena. The main toxins produced by the **dinoflagellate** *Alexandrium spp.* are **saxitoxins** and their analogues.

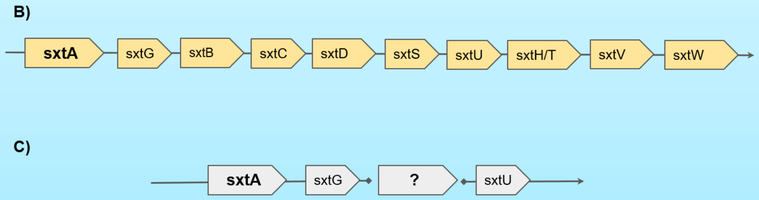
The marine dinoflagellate ***Alexandrium tamutum*** was discovered for the first time in the Gulf of Naples, and it is not known to produce saxitoxins. However, a clone of *A. tamutum* from the same Gulf showed toxicity on predators and anti-proliferative activity on human cells *[Lauritano et al., Front. Mar. Sci., 2016]*. A **RNA-seq-based pipeline** was used for *in silico* identification of transcripts that can be involved in the synthesis of toxic compounds (Both in control and **phosphate starvation** condition, to induce toxin production).

Alexandrium tamutum



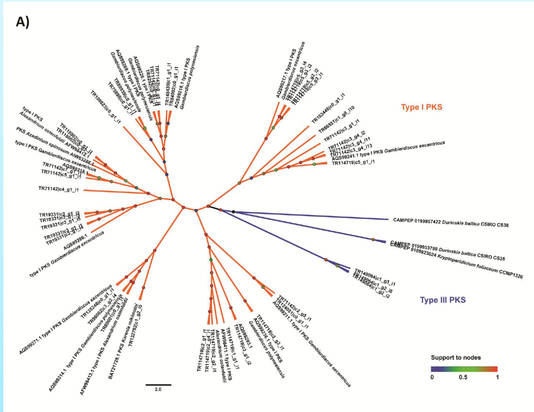
Using **Transcriptomic** data as its linchpin, this pipeline could further continue with the **heterologous expression** of the most interesting newfound enzymes, testing their activity with available or ad hoc assays. Finally both enzymes and produced toxins can be further characterized, in order to better evaluate their potential biotechnological application.

A. tamutum **transcriptome analysis** revealed the presence of many potential toxin-related transcripts **(**e.g**. 44 PKS genes**, distributed in the phylogenetic tree showed in **A)**) and new insights on still undiscovered pathways for toxins biosynthesis *[Vingiani et al. Marine Drugs, 2020]*. In fact, regarding saxitoxin pathway **only 18 sequences** related to sxtA, sxtG and sxtU genes were found, confirming the inability of this clone to produce saxitoxins (as showed in **B)** and **C)**).



B) Sxt biosynthetic pathway in dinoflagellates as reviewed by *Akbar* *[Akbar et al. Marine Drugs, 2020]* **C) sxt incomplete pathway as observed from *A. tamutum* transcriptome.**

3. Results



A) Phylogenetic tree of the newfound PKS genes in *A. tamutum*

RNA-seq approach strongly suggested the presence of toxic compounds, other than saxitoxins, which may be responsible for the toxic and antiproliferative activity of *A. tamutum*, as reported for other *Alexandrium spp.* which produced **less common** toxins.

These findings further indicated the chemistry of toxin production in dinoflagellates as **very complex and far from fully undiscovered**, requiring in depth new studies in both the ecological and drug discovery fields in order to identify novel chemical mediators in the marine environment and new compounds that can be potentially developed as pharmaceuticals.